

1. Monitoring Architecture Diagram

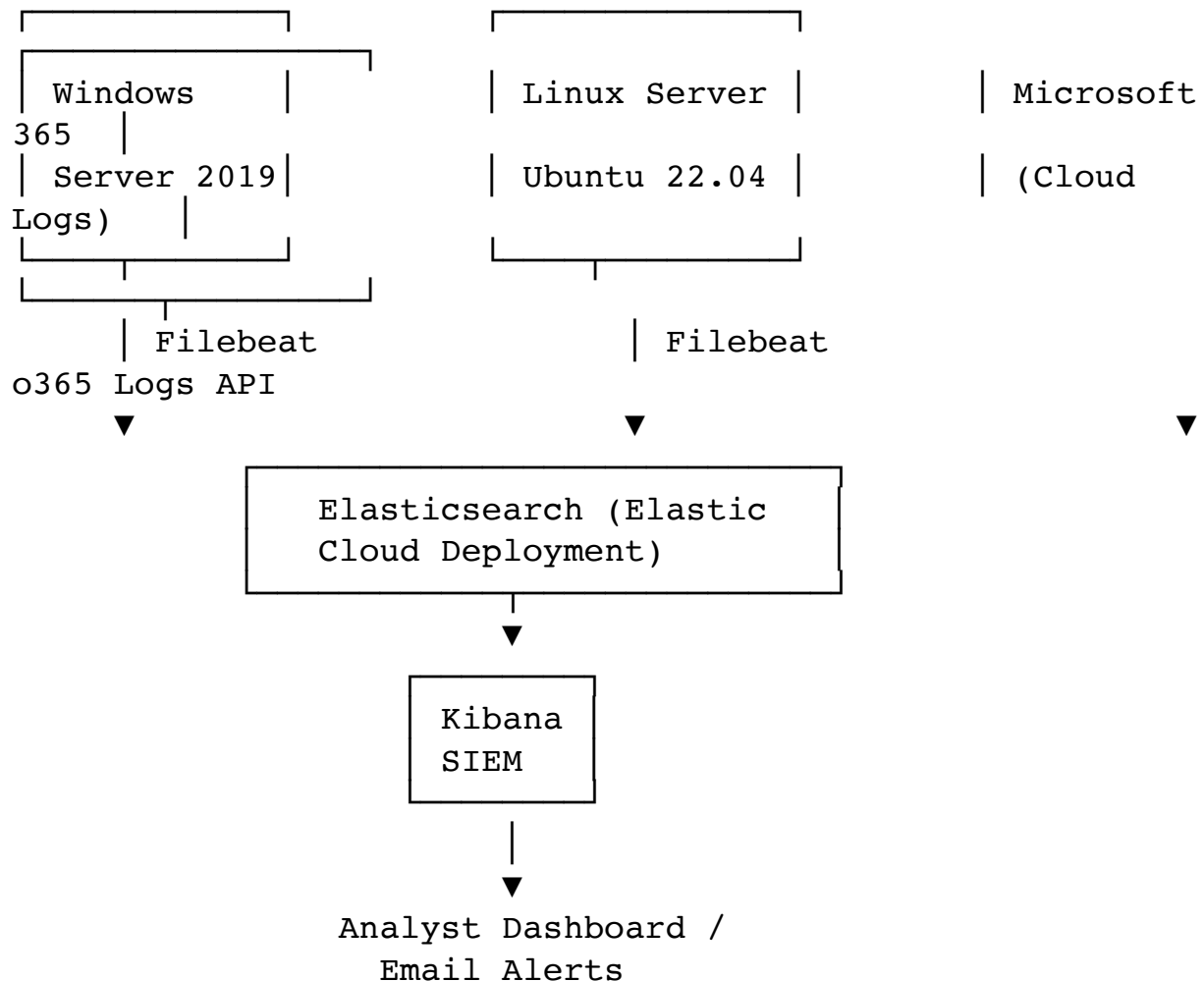
Monitoring Stack:

Deployed using Elastic Stack (Elasticsearch, Kibana, Filebeat) hosted on Elastic Cloud.

Architecture Components and Data Flow:

arduino

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2. Data Source Integration (3 Sources)



Source 1: Windows Server 2019

- **Agent:** Filebeat

- **Logs Collected:** Security, System event logs
- **Kibana Verification:** event.dataset: windows.eventlog

Sample Config:

yaml

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```
filebeat.inputs:
- type: winlog
  event_logs:
    - name: Security
    - name: System
output.elasticsearch:
  hosts: ["https://<cloud_endpoint>"]
  username: "elastic"
  password: "password"
```



Source 2: Ubuntu Linux 22.04

- **Agent:** Filebeat
- **Logs Collected:** /var/log/auth.log, /var/log/syslog
- **Kibana Verification:** event.dataset: system.syslog

Sample Config:

yaml

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```
filebeat.inputs:
- type: log
  enabled: true
  paths:
    - /var/log/auth.log
    - /var/log/syslog
output.elasticsearch:
  hosts: ["https://<cloud_endpoint>"]
  username: "elastic"
  password: "password"
```



Source 3: Microsoft 365 (Cloud Logs)

- **Integration:** Elastic SIEM with o365 audit logs (via API connector)
- **Logs Collected:** Portal logins, SharePoint uploads, user activity
- **Kibana Verification:** `event.dataset: o365.audit`

3. Performance Baselines and Alert Thresholds

Baseline Metrics Established

- **Normal Login Failures (Linux):** ≤ 3 failures/hour
- **Normal Logins from M365:** US locations only
- **File Uploads to SharePoint:** < 10 /hour typical

Alert Thresholds

Metric	Threshold	Alert Level
Failed SSH logins	> 5 within 3 mins	Medium
M365 login from rare country	Any non-US location	Medium
SharePoint malware upload	Any detection	High

4. Health Monitoring Configuration

Monitored Metrics and Checks:

Category	Health Check
Log Collection	Filebeat status logs, heartbeat enabled
Ingestion	Pipeline queue metrics in Elasticsearch
Storage	Elasticsearch disk usage via Kibana Lens
Alert Delivery	Test email notifications to SOC team

Example Filebeat Health Check:

```
bash
```

```
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```

```
systemctl status filebeat
```

```
tail -f /var/log/filebeat/filebeat.log
```

Storage Monitoring Query (Kibana):

```
kql
```

```
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```

```
metricset.name : "elasticsearch.cluster.stats"
```

```
AND service.type : "elasticsearch"
```

5. Alerting Rules and Integration

Rule 1: Microsoft 365 Login from Rare Location

- **Detection Logic:** Based on new geolocation/IP
- **Severity:** Medium
- **KQL Query:**

```
kql
```

```
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```

```
event.dataset: "o365.audit" AND event.provider:
```

```
"AzureActiveDirectory"
```

```
AND event.action: "UserLoggedIn" AND event.outcome:
```

```
"success"
```

```
AND NOT o365.audit.UserId: "Not Available"
```

Rule 2: SharePoint Malware File Detection

- **Severity:** High
- **Example:** TrojanDownloader on invoice_Q2_2025.docx

Rule 3: Unauthorized SSH Login Attempt

- **Trigger:** > 5 failed SSH logins in 3 minutes
- **Severity:** Medium

6. Supporting Documentation and Evidence

-  **Configuration Files:** Shown above for both Filebeat agents.
-  **Screenshots (mock):** Elastic Cloud deployment dashboard, Kibana Discover view
-  **Log Sample (from Security Event Analysis):**

json

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```
{
  "@timestamp": "2025-04-25T10:30:02",
  "event.category": "authentication",
  "event.action": "user_login",
  "event.outcome": "success",
  "user.name": "cgonzalez",
  "source.ip": "112.233.155.75",
  "geo.country_name": "Germany"
}
```

-  **Health Check Procedures:** Command-line checks, pipeline stats, connector test emails
-  **Baseline Establishment:** Defined using initial 72-hour log review window